Physics Multiple-Choice Questions

Isomorphic Question Set

Questions generated below. Multiple-choice answers in RED; incorrect answers in BLUE.

Q1. Which of the following modeling statements is most appropriate?

A. A bicycle can be treated as a point mass when analyzing the tension in one of its spokes during a turn.
This uses an oversimplified model when the detailed wheel geometry and spoke forces are exactly what matter.

B. A runner can be treated as a perfectly rigid block when analyzing the motion of the runner's knee joint.
This ignores the internal limb motion that is central to a joint-level analysis.

C. A research balloon can be treated as a point mass when tracking its change in position over a 25 km flight.
This is correct because the flight distance is much larger than the balloon's size, so only its translational motion matters. In that approximation, the balloon's shape is irrelevant and its motion can be modeled using a single position vector for a point mass, $\vec{r}(t)$.

D. If air resistance is neglected in a calculation, it is never acceptable to model the object as moving in a vacuum.
This mistakes a useful idealization for a literal claim; assuming a vacuum is often the standard way to set $F_{\text{drag}} = 0$.

Answer: *C*

Q2. Which of the following unit equalities is correct?

A. $1\,\text{ms} = 1000\,\text{s}$
This reverses the milli-prefix; milli $= 10^{-3}$, not 10^{3}.

B. $1\,\text{g} = 10^{-3}\,\text{kg}$
This is correct because the prefix relation is $1\,\text{kg} = 1000\,\text{g}$. Dividing both sides by 1000 gives $1\,\text{g} = 10^{-3}\,\text{kg}$, so the two expressions represent the same mass in different units.

C. $1\,\text{cm} = 0.1\,\text{m}$
This is a decimal-place error; $1\,\text{cm} = 10^{-2}\,\text{m}$, not $10^{-1}\,\text{m}$.

D. $1000\,\text{g} = 10^{-6}\,\text{kg}$
This applies the conversion factor incorrectly; 1000 g equals 1 kg.

Answer: *B*

Q3. Which option lists an incorrect SI unit for the quantity named?

A. Impulse: $\text{N} \cdot \text{s}$
This is a correct unit because impulse equals change in momentum, and $\text{N} \cdot \text{s} = \text{kg} \cdot \text{m/s}$.

B. Density: kg/m^3
This is the standard SI unit for mass density, so choosing it would reject a valid derived unit.

C. Work: N/m
This is incorrect because work has dimensions of force times distance, $W = F\Delta x$, so its SI unit is $\text{N} \cdot \text{m} = \text{J}$. Writing N/m would instead describe force per length, which is a different physical quantity.

D. Frequency: s^{-1}
This is correct because the hertz is defined as 1/s.

Answer: *C*

Q4. A survey drone is located at position $\vec{r} = -8\hat{\imath} + 15\hat{\jmath}$ meters relative to its launch point at the origin. Which statement is correct?

A. The drone is 23 m from the launch point.
This adds the components directly instead of taking the magnitude of the position vector.

B. The drone is 15 m from the launch point.
This uses only the y-component and ignores the displacement in the x-direction.

C. The drone is 17 m from the launch point.
This is correct because the distance from the origin is the magnitude of the position vector, $r = \sqrt{(-8)^2 + 15^2}$. Evaluating gives $r = \sqrt{64 + 225} = \sqrt{289} = 17\,\text{m}$, not the sum or an individual component.

D. The drone is 8 m from the launch point.
This uses only the magnitude of the x-component and does not account for the full two-dimensional displacement.

Answer: *C*

Q5. If the position of a particle is given by $\vec{s}(t) = \left[(0.90\,\text{m/s}^3)t^3 + (1.80\,\text{m/s}^2)t^2\right]\hat{\imath}$, which statement is correct?

A. The motion has constant acceleration.
This ignores that differentiating a cubic position function produces an acceleration that depends on t.

B. The magnitude of the acceleration is 1.80 m/s^2 at all times.
This confuses a coefficient in the position function with the acceleration obtained after taking two derivatives.

C. The magnitude of the acceleration at $t = 0$ is 3.60 m/s^2.
This is correct because $\vec{v}(t) = \dfrac{d\vec{s}}{dt} = \left[(2.70)t^2 + (3.60)t\right]\hat{\imath}$ and $\vec{a}(t) = \dfrac{d\vec{v}}{dt} = \left[(5.40)t + 3.60\right]\hat{\imath}$. Substituting $t = 0$ gives $\vec{a}(0) = 3.60\hat{\imath}\,\text{m/s}^2$, so the initial acceleration magnitude is $3.60\,\text{m/s}^2$.

D. The magnitude of the acceleration at $t = 0$ is 0 m/s^2.
This misses the nonzero constant term that remains after differentiating twice.

Answer: *C*

Q6. An object moves along a straight path from position $\vec{r}_A = 1\hat{\imath} - 2\hat{\jmath}\,\text{m}$ to position $\vec{r}_B = 13\hat{\imath} + 3\hat{\jmath}\,\text{m}$ in 4.0 s. Which statement is correct?

A. The distance traveled is 13 m.
This is correct because the path is stated to be straight, so the travel distance equals the magnitude of the displacement vector. Here $\Delta\vec{r} = (13 - 1)\hat{\imath} + [3 - (-2)]\hat{\jmath} = 12\hat{\imath} + 5\hat{\jmath}$, so the distance is $\sqrt{12^2 + 5^2} = 13\,\text{m}$.

B. The speed at $t = 0$ must be 3.25 m/s.
This treats the average speed, 13/4, as though it must equal the instantaneous speed at the initial moment.

C. The acceleration at $t = 0$ must be 0.81 m/s^2.
This invents a unique acceleration from distance and elapsed time even though many different straight-line motions are possible.

D. None of the above is correct.
This overlooks that the straight-line distance between the two stated positions is exactly 13 m.

Answer: *A*

Q7. A frog leaps from a lily pad with launch speed v_0 at an angle θ above the horizontal. Neglect air resistance. What is the frog's speed at the highest point of its trajectory?

A. 0
This assumes the frog stops completely at the top, ignoring its horizontal motion.

B. $v_0 \sin\theta$
This is the initial vertical component, but that component becomes zero at maximum height.

C. $v_0 \cos\theta$
This is correct because in projectile motion without air resistance, the horizontal component of velocity remains constant while the vertical component is zero at the top. Therefore the speed there is just the horizontal component, $v = v_0 \cos\theta$.

D. v_0
This incorrectly assumes the full launch speed is unchanged even though the vertical component has been reduced to zero at the highest point.

Answer: *C*

Q8. A warehouse robot initially moves with velocity $\vec{v}_0 = (12.0\,\text{m/s})\hat{\imath}$. Its side thrusters then produce a constant acceleration $\vec{a} = (-1.50\,\text{m/s}^2)\hat{\imath} + (2.50\,\text{m/s}^2)\hat{\jmath}$. Which statement is correct?

A. The x-component of the robot's velocity remains constant.
This ignores the nonzero x-acceleration, so v_x must change with time.

B. The y-component of the robot's velocity remains constant.
This ignores the nonzero y-acceleration, which makes v_y increase from its initial value of zero.

C. Both the direction of motion and the speed change.
This is correct because the acceleration changes both components of velocity: $v_x(t) = 12.0 - 1.50t$ and $v_y(t) = 2.50t$. Since the vector components change, the direction changes, and because the magnitude $v = \sqrt{v_x^2 + v_y^2}$ also varies with time, the speed changes as well.

D. None of the above is correct.
This misses that a two-component acceleration changes the velocity vector in both magnitude and direction.

Answer: *C*

Q9. Two point charges are both doubled in magnitude, and the distance between them is cut in half. What happens to the magnitude of the electrostatic force between them?

A. It stays the same.
This ignores that both the charge product and the inverse-square distance factor change.

B. It decreases by a factor of 4.
This reverses the effect of reducing the separation; bringing charges closer increases the force.

C. It increases by a factor of 4.
This accounts for doubling both charges but misses the additional factor from halving the distance.

D. It increases by a factor of 8.
This reflects incomplete scaling of Coulomb's law and neglects part of the inverse-square dependence.

E. It increases by a factor of 16.
This is correct from Coulomb's law, $F = k\dfrac{|q_1 q_2|}{r^2}$. Doubling both charges multiplies the numerator by $2 \times 2 = 4$, and halving the separation multiplies the force by another factor of $1/(1/2)^2 = 4$, so the total change is $4 \times 4 = 16$.

Answer: *E*

Q10. A probe is launched from the surface of a spherical moon. The escape speed is $v_e = \sqrt{\dfrac{2GM}{R}}$, where M and R are the moon's mass and radius. Which statement is correct?

A. A moon with greater mass and smaller radius requires a larger escape speed.
This is correct because $v_e \propto \sqrt{M/R}$. Increasing M raises the quantity under the square root, and decreasing R also raises it, so both changes increase the required escape speed.

B. A heavier probe requires a larger escape speed than a lighter probe from the same moon.
This confuses the mass of the escaping object with the parameters in the formula; the probe's mass does not appear in v_e.

C. A moon with smaller mass and larger radius requires a larger escape speed.
This reverses both dependencies, since smaller M and larger R each reduce v_e.

D. A lighter probe requires a smaller escape speed from the same moon.
This repeats the misconception that escape speed depends on the mass of the object trying to escape.

Answer: *A*

Q11. A constant force $\vec{F} = (5.20\,\text{N})\hat{\imath} + (-2.40\,\text{N})\hat{\jmath}$ acts on a particle as it moves along the y-axis from $y = -1.0\,\text{m}$ to $y = 3.5\,\text{m}$. What work is done by the force on the particle?

A. $-10.8\,\text{J}$
This is correct because work is the dot product $W = \vec{F} \cdot \Delta\vec{r}$. The displacement is only along y, with $\Delta\vec{r} = (0\hat{\imath} + 4.5\hat{\jmath})\,\text{m}$, so $W = (5.20)(0) + (-2.40)(4.5) = -10.8\,\text{J}$.

B. $23.4\,\text{J}$
This uses the x-component of force even though there is no displacement in the x-direction.

C. $12.0\,\text{J}$
This likely uses only the magnitude of the y-force and misses the negative sign from the opposite direction of force and displacement.

D. $28.8\,\text{J}$
This appears to combine both force components as if all of them contributed to work along the actual displacement.

Answer: *A*

Q12. A $1.6\,\text{kg}$ cart moves at $4.0\,\text{m/s}$ toward a wall, rebounds at $1.5\,\text{m/s}$ in the opposite direction, and remains in contact with the wall for $0.20\,\text{s}$. What is the magnitude of the average force on the cart during the collision?

A. $20\,\text{N}$
This underestimates the force by not fully accounting for the reversal in velocity during the collision.

B. $28\,\text{N}$
This likely uses an incorrect change in speed or omits part of the impulse calculation.

C. $44\,\text{N}$
This is correct from the impulse-momentum theorem, $F_{\text{avg}} = \dfrac{m(v_f - v_i)}{\Delta t}$. Taking the initial direction as positive gives $v_i = 4.0\,\text{m/s}$ and $v_f = -1.5\,\text{m/s}$, so $F_{\text{avg}} = \dfrac{1.6(-1.5 - 4.0)}{0.20} = -44\,\text{N}$; the magnitude is $44\,\text{N}$.

D. None of the above is correct.
This ignores that the impulse calculation gives a force magnitude of $44\,\text{N}$.

Answer: *C*

Q13. A 65 kg skater is initially at rest on frictionless ice. A 0.25 kg training bag is thrown horizontally at 54 km/h, and the skater catches it. What is the skater's speed immediately after the catch?

A. 0.057 m/s
This is correct by conservation of momentum for a perfectly inelastic catch: $m_b v_b = (m_s + m_b)v_f$. Converting 54 km/h to 15 m/s gives $v_f = \dfrac{(0.25)(15)}{65.25} \approx 0.057$ m/s.

B. 0.24 m/s
This is too large and usually comes from neglecting the combined final mass after the catch.

C. 3.75 m/s
This treats the bag's momentum mv as though it were the final speed, ignoring mass division and momentum sharing.

D. 15 m/s
This assumes the skater moves away with the bag's original speed, which violates momentum conservation for a much more massive catcher.

Answer: *A*
